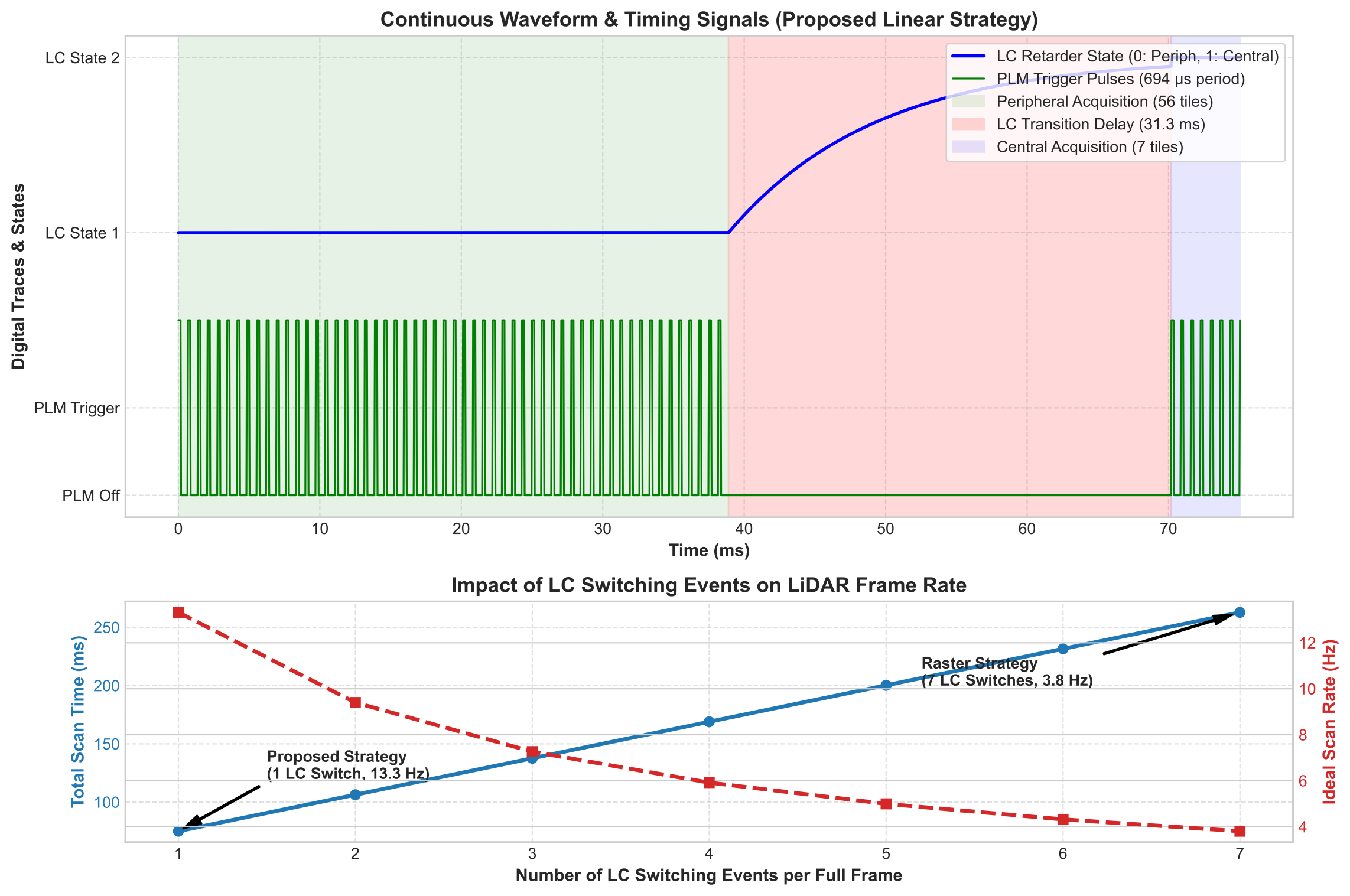
Continuous Waveform & Timing Signals (Proposed Linear Strategy)
LC State 2
LC State 1
PLM Trigger
PLM Off
Digital Traces & States
LC Retarder State (0: Periph, 1: Central)
PLM Trigger Pulses (694 µs period)
Peripheral Acquisition (56 tiles)
LC Transition Delay (31.3 ms)
Central Acquisition (7 tiles)
0
10
20
30
40
50
60
70
Time (ms)
Impact of LC Switching Events on LiDAR Frame Rate
Total Scan Time (ms)
Ideal Scan Rate (Hz)
Proposed Strategy
(1 LC Switch, 13.3 Hz)
Raster Strategy
(7 LC Switches, 3.8 Hz)
Number of LC Switching Events per Full Frame